

VOTING MACHINE

Using AVR Microcontroller

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Polling by electronic voting machines (EVMs) has become a norm during elections. Unlike paper ballot systems, EVMs are safe and reliable as these rule out the possibility of invalid voting.

These are more economical and efficient too. Their compact size saves transportation costs.

Here we describe the design of an electronic voting machine based on ATmega16A microcontroller. To cast a vote, users need to press the key against the name of the candidate of their choice. The vote is automatically saved in the microcontroller, and simultaneously the LCD shows the message "Thank You." The author's prototype is shown in Fig. 1.

Circuit and working

As shown in Fig. 2, the circuit of the electronic voting machine is built around ATmega16A microcontroller (IC2), IC 7805 (IC1), a 16×2 LCD (LCD1) and other associated components.

The controller used here operates off 5V DC supply. So a 7805 voltage regulator IC is used to step 9V DC down to 5V. Capacitors C1 and C2 are used to reduce ripples. ATmega16A has 32 general purpose input/output (I/O) pins. An external 8MHz crystal oscillator is used to provide the timing pulse. Four switches (S1 through S4) are connected to Port B (PB0, PB1, PB2 and PB3) of IC2.

The LCD (LCD1) operates in 4-bit



Fig. 1: Author's prototype

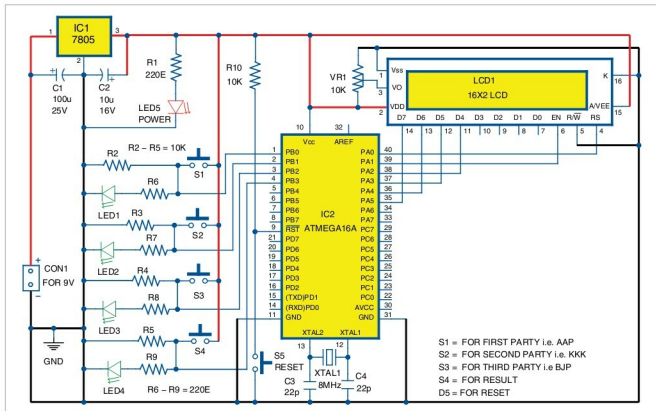


Fig. 2: Circuit diagram of the electronic voting machine using ATmega16A microcontroller